

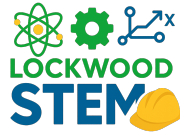
Introduction to Engineering Design

Aerospace Academy Version

Unit 3 Rubric

Introduction to Engineering Design

Course: Introduction to Engineering Design
Program: Academy of Advanced Aerospace Engineering
Brand: Lockwood STEM
Use: Student-facing and teacher grading rubrics



Unit 3

CAD Modeling for Aerospace Components

Fusion workflow, parametric sketches, constraints, features, inspection, and aerospace component modeling project.

Grading Rubrics and Scoring Guide

Unit 3: CAD Modeling for Aerospace Components

Assessment Purpose

Fusion workflow, parametric sketches, constraints, features, inspection, and aerospace component modeling project. This rubric is designed for consistent grading, feedback, and student self-checks.

Suggested 4-Point Scale

4 = Expert: exceeds expectations with clear evidence, precision, and independence. **3 = Proficient:** meets expectations with complete and accurate work. **2 = Emerging:** partially meets expectations; work is incomplete, vague, or inconsistent. **1 = Incomplete:** missing, unsafe, inaccurate, or shows limited evidence of understanding.

Unit 3 Score Summary

Category	Points Possible	Points Earned
File setup and digital workflow	10	
Sketch constraints and dimensions	15	
Parametric design intent	15	
Core CAD features	15	
Aerospace component function	15	
Model accuracy and inspection	10	
Documentation and screenshots	10	
Iteration and improvement	10	

Unit 3 Project Rubric



Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete	
File Setup & Digital Workflow	10	Files are correctly named, organized, versioned, and submitted with required exports/screenshots.	Files are mostly correct with minor naming/organization issues.	Multiple file errors or missing exports.	Files missing or unusable.	
Sketch Constraints & Dimensions	15	Sketches are fully or appropriately constrained and driven by intentional dimensions.	Sketches are mostly constrained with minor issues.	Sketches rely on loose geometry or missing dimensions.	Sketches are unconstrained or incorrect.	
Parametric Design Intent	15	Model uses parameters/features that make design changes predictable and efficient.	Model shows mostly logical feature order and editable dimensions.	Model is difficult to edit or lacks clear design intent.	Model is not parametric or cannot be revised.	
Core CAD Features	15	Uses extrude, revolve/sweep as needed, fillet/chamfer, shell/hole/pattern tools appropriately.	Uses required core features correctly with minor errors.	Uses limited features or features are applied incorrectly.	Required CAD features missing.	
Aerospace Component Function	15	Component is clearly connected to an aerospace purpose and considers payload, mounting, weight, clearance, or manufacturing.	Component has a reasonable aerospace connection.	Aerospace purpose is weak or mostly decorative.	No clear aerospace function.	
Model Accuracy & Inspection	10	Dimensions, fit, mass/size, clearances, and errors are checked and documented.	Basic inspection completed.	Inspection is incomplete or superficial.	No inspection evidence.	
Documentation & Screenshots	10	Includes clean views, feature explanation, design notes, and any required export files.	Documentation is adequate.	Documentation is incomplete or hard to follow.	Missing.	
Iteration & Improvement	10	Shows meaningful revisions based on testing, feedback, or identified issues.	Shows at least one useful revision.	Revision is minor or not explained.	No revision evidence.	
						Rubrics

Student Self-Check Before Submission

	Requirement	Complete?
☐	File setup and digital workflow evidence is complete and ready to be graded.	
☐	Sketch constraints and dimensions evidence is complete and ready to be graded.	
☐	Parametric design intent evidence is complete and ready to be graded.	
☐	Core CAD features evidence is complete and ready to be graded.	
☐	Aerospace component function evidence is complete and ready to be graded.	
☐	Model accuracy and inspection evidence is complete and ready to be graded.	
☐	Documentation and screenshots evidence is complete and ready to be graded.	
☐	Iteration and improvement evidence is complete and ready to be graded.	
☐	All pages/files are named, organized, and submitted correctly.	
☐	Reflection or revision notes are complete.	

Teacher
Use the
evidence
those b